

Deterministic Optimization

Large-Scale Optimization

Dantzig-Wolfe
Decomposition

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Dantzig-Wolfe Decomposition 3

Large-Scale Optimization

Dantzig-Wolfe

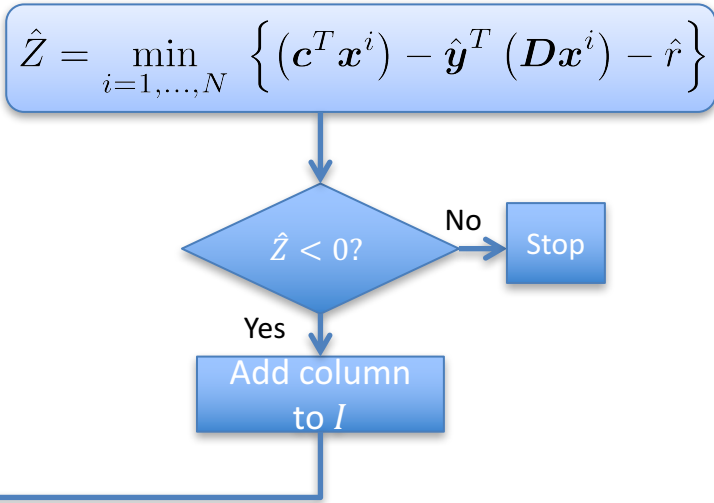
Learning Objectives for this lesson

- Develop the last step in the Dantzig-Wolfe decomposition algorithm

Dantzig-Wolfe Decomposition

- Step 4: Apply Column Generation to solve (MP)
 - Check optimality of $\hat{\lambda}$ using dual solution $\hat{y}, \hat{r}$:

$$\begin{aligned} (RMP) \quad & \min_{\lambda_1, \dots, \lambda_k} \begin{bmatrix} \mathbf{c}^T \mathbf{x}^1 & \mathbf{c}^T \mathbf{x}^2 & \dots & \mathbf{c}^T \mathbf{x}^k \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_k \end{bmatrix} \\ \text{s.t.} \quad & \begin{bmatrix} \mathbf{D}\mathbf{x}^1 & \mathbf{D}\mathbf{x}^2 & \dots & \mathbf{D}\mathbf{x}^k \\ 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_k \end{bmatrix} = \begin{bmatrix} \mathbf{b}_0 \\ 1 \end{bmatrix} \\ & \lambda_i \geq 0, \forall i \in I = \{1, \dots, k\}. \end{aligned}$$



How To Compute Smallest Reduced Cost?

- Compute smallest reduced cost: $\hat{Z} = \min_{i=1,\dots,N} \left\{ (c^T x^i) - \hat{y}^T (Dx^i) - \hat{r} \right\}$
 - We don't want to enumerate over all extreme points:
- Key idea: Reverse use of extreme point representation

$$\hat{Z} = \min_{i=1,\dots,N} \left\{ (c^T x^i) - \hat{y}^T (Dx^i) - \hat{r} \right\}$$

$$= \min_x \left\{ (c^T x) - \hat{y}^T (Dx) - \hat{r} \right\}$$

s.t. $x \in P$

$$= \min_x \left\{ (c^T - \hat{y}^T D) x \right\} - \hat{r}$$

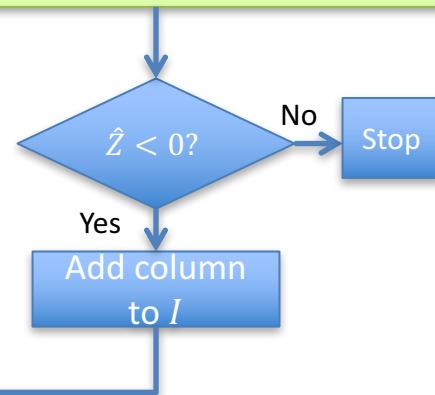
s.t. $Fx = b, x \geq 0$

Dantzig-Wolfe Decomposition

- Start with a subset of columns I to form (RMP)
- Solve (RMP) and get optimal dual variable $\hat{y}, \hat{r}$
- Check optimality of (RMP)'s primal solution

$$\begin{aligned} (RMP) \quad & \min_{\lambda_1, \dots, \lambda_k} \begin{bmatrix} \mathbf{c}^T \mathbf{x}^1 & \mathbf{c}^T \mathbf{x}^2 & \dots & \mathbf{c}^T \mathbf{x}^k \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_k \end{bmatrix} \\ \text{s.t.} \quad & \begin{bmatrix} \mathbf{D}\mathbf{x}^1 & \mathbf{D}\mathbf{x}^2 & \dots & \mathbf{D}\mathbf{x}^k \\ 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_k \end{bmatrix} = \begin{bmatrix} \mathbf{b}_0 \\ 1 \end{bmatrix} \\ & \lambda_i \geq 0, \forall i \in I = \{1, \dots, k\}. \end{aligned}$$

$$\begin{aligned} \hat{Z} = \min_{\mathbf{x}} \quad & \left\{ \left(\mathbf{c}^T - \hat{\mathbf{y}}^T \mathbf{D} \right) \mathbf{x} \right\} - \hat{r} \\ \text{s.t.} \quad & \mathbf{F}\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0} \end{aligned}$$



Summary

- We learned:
 - The fourth step in Dantzig-Wolfe decomposition that applies column generation to solve the transformed master problem